R Sumanth

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.09
KENDRIYA VIDYALAYA SHIVRAMPALLY HYDERABAD TL	Class XII (CBSE)	2024	67.00%
KENDRIYA VIDYALAYA SHIVRAMPALLY HYDERABAD TL	Class X (CBSE)	2022	75.40%

RELEVANT COURSES

AI: Foundations of Machine Learning | Deep Learning | AI for Humanity | Introduction to Modern AI | Introduction to Classical AI | Programming for AI

CS: Data Structures & Applications | Algorithms | DBMS | Discrete Mathematics | Introduction to Programming |

Mathematics: Probability and Random Variables | Optimization I & II | Numerical Analysis | Concentration Inequalities | Metric Spaces | Vector Calculus | Differential Equations | Series of Function | Calculus

EE: Control Systems | Matrix Theory

PROJECTS

Mini Machine Learning Library & Autograd Engine

- Built a **scikit-learn-style ML library** and **PyTorch-inspired autograd engine** from scratch using Python and NumPy.
- Implemented **Logistic Regression, neural networks, preprocessing utilities, and optimization algorithms**.
- Developed modules for **feature engineering, K-Fold cross-validation, and gradient-based training**.
- Trained and optimized models on **Fashion-MNIST** using hyperparameter tuning and model evaluation.

Deep Learning Systems & Sequence Models Implementation

- Developed **CNN architectures** in PyTorch for image classification and evaluated multiple optimization techniques.
- Implemented **RNN, LSTM, Transformer, and State Space Models (SSMs)** for sequence learning tasks.
- Compared models using **Accuracy, Precision, Recall, and F1-score** across different datasets.
- Tuned model architectures and hyperparameters to improve convergence and generalization.

Synthetic Shape Classification using Multi-Scale Residual CNN

- Developed a **modular PyTorch pipeline** for binary image classification using a **custom Residual CNN** with multi-scale training (96×96, 128×128, 160×160).
- Improved model robustness through **Stratified 5-Fold Cross-Validation, Test-Time Augmentation (TTA), threshold optimization, and confidence-based prediction calibration**.
- Built reusable modules for **data loading, preprocessing, model training, evaluation, and metric computation**, ensuring a reproducible deep learning workflow.
- Achieved **~99% out-of-fold validation accuracy** and analyzed the generalization gap using comprehensive evaluation and calibration techniques.

Exploratory Data Analysis & Insights on Real-World Data

- Performed **data cleaning, preprocessing, and exploratory data analysis (EDA)** on real-world datasets.
- Analyzed **feature distributions, correlations, missing values, and outliers** using statistical techniques.
- Built visualizations with **Pandas, NumPy, and Matplotlib** to uncover patterns and trends.
- Generated actionable insights to support **data-driven decision-making and model development**.

SKILLS

Programming Languages: Python, C, C++, SQL, HTML, CSS

Machine Learning & AI: Machine Learning, Deep Learning

Libraries & Frameworks: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, SciPy

Developer Tools: Git, GitHub, Linux, VS Code, Jupyter Notebook, Google Colab, MATLAB, Excel

Core CS: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems

Soft Skills: Problem Solving, Analytical Thinking, Critical Thinking, Teamwork, Leadership, Communication, Time Management, Adaptability, Quick Learning, Attention to Detail

EXTRACURRICULAR

- Awarded **Bronze Medal** in **Kabaddi** at the **Milan 2025 Inter-Hostel Competition**, IIT Hyderabad.
- Awarded **Bronze Medal** in **Kabaddi** at the **Diesta Inter-Departmental Competition**, IIT Hyderabad